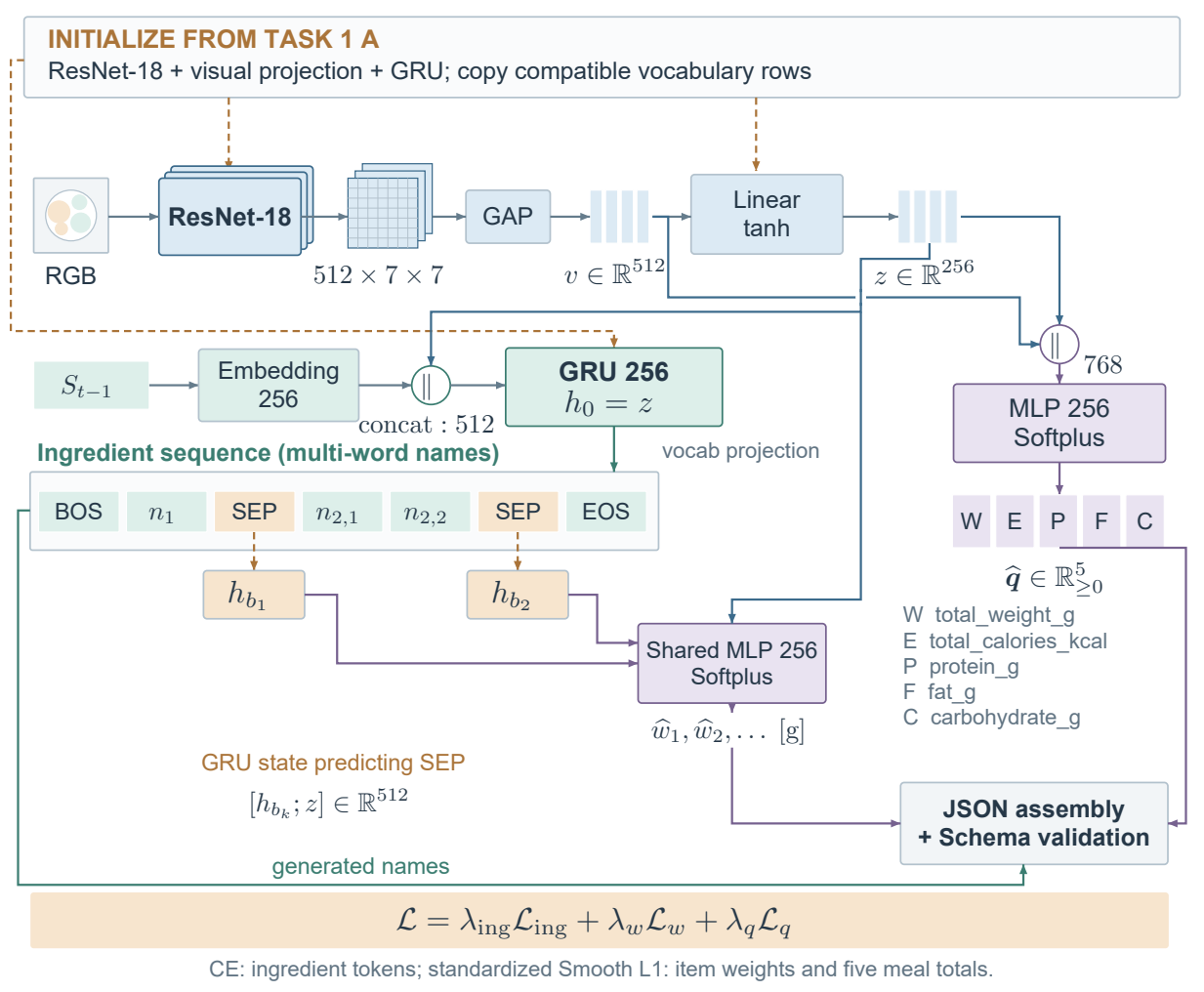

INITIALIZE FROM TASK 1 A
ResNet-18 + visual projection + GRU; copy compatible vocabulary rows
RGB
ResNet-18
$512 \times 7 \times 7$
GAP
$v \in \mathbb{R}^{512}$
Linear tanh
$z \in \mathbb{R}^{256}$
768
S_{t-1}
Embedding 256
concat : 512
GRU 256
$h_0 = z$
vocab projection
MLP 256 Softplus
Ingredient sequence (multi-word names)
BOS
n_1
SEP
$n_{2,1}$
$n_{2,2}$
SEP
EOS
W
E
P
F
C
h_{b_1}
h_{b_2}
$\widehat{q} \in \mathbb{R}^5_{\geq 0}$
W total_weight_g
E total_calories_kcal
P protein_g
F fat_g
C carbohydrate_g
Shared MLP 256 Softplus
$\widehat{w}_1, \widehat{w}_2, \ldots$ [g]
GRU state predicting SEP
$[h_{b_k}; z] \in \mathbb{R}^{512}$
JSON assembly + Schema validation
generated names
$\mathcal{L} = \lambda_{\text{ing}}\mathcal{L}_{\text{ing}} + \lambda_w\mathcal{L}_w + \lambda_q\mathcal{L}_q$
CE: ingredient tokens; standardized Smooth L1: item weights and five meal totals.